# CH1101 : Quantum Chemistry & Spectroscopy

Lec 10

15 | 09 | 2026

## Motion in 2-Dimension (x, y)

TISE

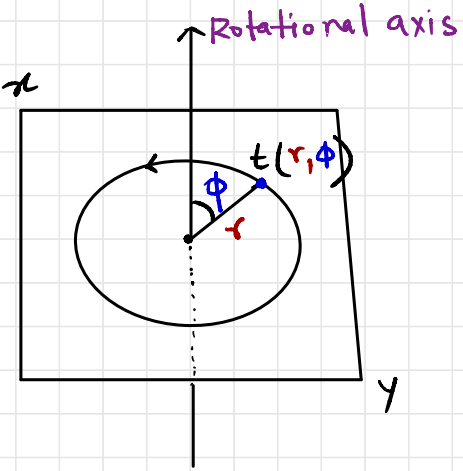
$$-\frac{\hbar^2}{2m}\left[\frac{\partial^2\psi(x,y)}{\partial x^2}+\frac{\partial^2\psi(x,y)}{\partial y^2}\right]=E\psi(x,y)$$

### Constrained free-particle

### Rotational motion in a ring

$\phi$ : Rotational angle

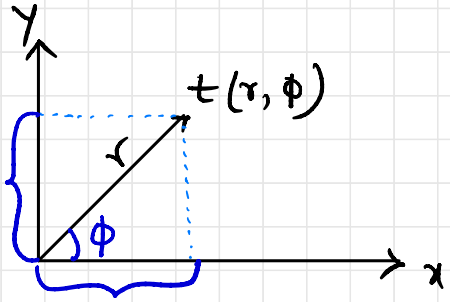
$r$ : Radius of the ring which is placed on a plane




Coordinate transformation to get TISE from Cartesian to Polar (plane) coordinate system

$$x = r\cos\phi$$

$$y = r\sin\phi$$



TISE in 2D

$$-\frac{\hbar^2}{2m}\left[\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}\right]=E\psi$$

1. Apply coordinate transformation
2. Separation of variables

$V = 0$ As it's a free particle

$$\Rightarrow -\frac{\hbar^2}{2I}\frac{d^2\psi}{d\phi^2}=E\psi$$

$I$: moment of inertia $= mr^2$

$$\Rightarrow -\frac{d^2\psi}{d\phi^2}=\frac{2IE}{\hbar^2}\psi$$

$$m_l^2=\frac{2IE}{\hbar^2}$$

$$\Rightarrow \frac{d^2\psi}{d\phi^2}+m_l^2\psi=0$$

**Wavefunction** $\psi(\phi)=Ae^{im_l\phi}+Be^{-im_l\phi}$

**Energy** $m_l^2=\frac{2IE}{\hbar^2}$

$$\Rightarrow E=\frac{m_l^2\hbar^2}{2I}$$

$$=\frac{m_l^2\hbar^2}{8\pi^2mr^2}$$

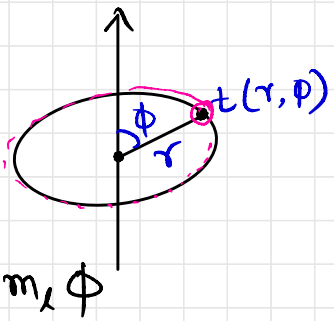
$A, B, m_l$ are yet to be determined!

by substituting $I = mr^2$

$m_l$, A & B yet to be derived/defined!

## Cyclic Boundary Conditions:

$$\psi(\phi) = \psi(\phi + 2\pi)$$




Wavefunction

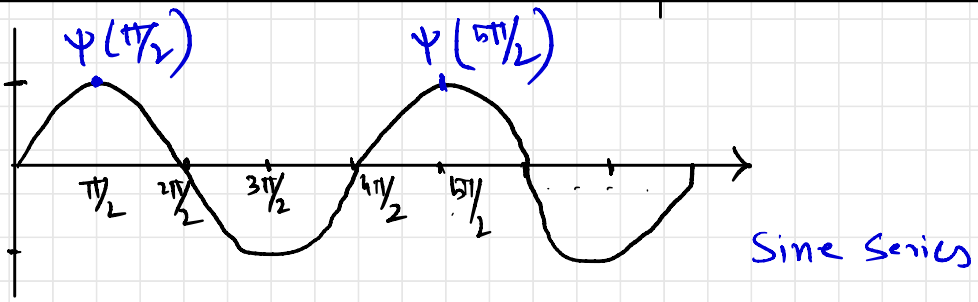
$$\psi(\phi) = A e^{i m_l \phi} + B e^{-i m_l \phi}$$

Apply BC

$$\psi(\phi) = C e^{i m_l \phi}$$

↑ Another constant

For $m_l = 0, \pm 1, \pm 2, \pm 3, \ldots$




Sine Series

$$A e^{i m_l \phi} + B e^{-i m_l \phi}$$

Applying $B e^{\cdots}$

$$\Rightarrow A e^{i m_l (\phi + 2\pi)} + B e^{-i m_l (\phi + 2\pi)}$$

$$\Rightarrow A e^{i m_l \phi} \times \underbrace{e^{i m_l 2\pi}}_{1} + B e^{-i m_l \phi} \times \underbrace{e^{-i m_l 2\pi}}_{1}$$

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NOTE $\quad e^{i m_l 2\pi} = \cos(m_l 2\pi) + i \sin(m_l 2\pi)$

$= 1 + 0 \quad$ for $m_l = 0, \pm 1, \pm 2, \ldots$

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$$\Rightarrow A e^{i m_l \phi} + B e^{-i m_l \phi}$$

$$\Rightarrow \underbrace{(A+B)}\, e^{i m_l \phi} \Rightarrow C e^{i m_l \phi}$$

$$\boxed{\psi(\phi) = C e^{i m_l \phi}}$$

Find $C$ constant

$$P = \int_0^{2\pi} \psi^* \psi \, d\phi = 1$$

Apply Normalization condition

$$C = \frac{1}{\sqrt{2\pi}}$$

Final wavefunction

$$\boxed{\psi(\phi) = \frac{1}{\sqrt{2\pi}} e^{i m_l \phi}}$$

$m_l = 0, \pm 1, \pm 2 \ldots$